

PHYSIVERSE

Phase 4 – Database Design & Architecture

Version: 1.0

Database Philosophy

The database should be:

- Scalable
 - Secure
 - Modular
 - Analytics Ready
 - AI Ready
 - Real-Time Ready
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Technology Stack

Database: PostgreSQL

Backend: Supabase

Authentication: Supabase Auth

Storage: Supabase Storage

Realtime: Supabase Realtime

Caching: Redis (Future)

High Level Architecture

Authentication

↓

User Profiles

↓

Learning System
↓
Simulations & Labs
↓
Gamification
↓
Community
↓
Analytics
↓
AI Services

Authentication Tables

users

Field	Type
id	UUID
email	TEXT
created_at	TIMESTAMP
last_login	TIMESTAMP

profiles

Field	Type
id	UUID
full_name	TEXT
avatar	TEXT
bio	TEXT
role	TEXT
class_level	TEXT
country	TEXT
created_at	TIMESTAMP

roles

Field	Type
id	UUID
name	TEXT
permissions	JSON

Learning System

subjects

Field	Type
id	UUID
title	TEXT
description	TEXT
icon	TEXT

units

Field	Type
id	UUID
subject_id	UUID
title	TEXT

chapters

Field	Type
id	UUID
unit_id	UUID
title	TEXT

Field	Type
order	INTEGER

lessons

Field	Type
id	UUID
chapter_id	UUID
title	TEXT
slug	TEXT
content	JSONB
estimated_time	INTEGER

topics

Field	Type
id	UUID
lesson_id	UUID
title	TEXT
type	TEXT

Simulations System

simulations

Field	Type
id	UUID
title	TEXT
description	TEXT
category_id	UUID

Field	Type
difficulty	TEXT
thumbnail	TEXT

simulation_categories

Field	Type
id	UUID
title	TEXT

simulation_progress

Field	Type
id	UUID
user_id	UUID
simulation_id	UUID
completion	INTEGER
score	INTEGER

Virtual Labs

labs

Field	Type
id	UUID
title	TEXT
description	TEXT

lab_sessions

Field	Type
id	UUID
user_id	UUID
lab_id	UUID
observations	JSONB
results	JSONB

Quiz System

quizzes

Field	Type
id	UUID
lesson_id	UUID
title	TEXT

questions

Field	Type
id	UUID
quiz_id	UUID
type	TEXT
question	TEXT
explanation	TEXT

answers

Field	Type
id	UUID
question_id	UUID
answer	TEXT
is_correct	BOOLEAN

quiz_attempts

Field	Type
id	UUID
user_id	UUID
quiz_id	UUID
score	INTEGER
completed_at	TIMESTAMP

Progress Tracking

lesson_progress

Field	Type
user_id	UUID
lesson_id	UUID
completion	INTEGER

chapter_progress

Field	Type
user_id	UUID

Field	Type
chapter_id	UUID
completion	INTEGER

course_progress

Field	Type
user_id	UUID
subject_id	UUID
completion	INTEGER

Notes System

notes

Field	Type
id	UUID
user_id	UUID
lesson_id	UUID
content	TEXT

bookmarks

Field	Type
id	UUID
user_id	UUID
lesson_id	UUID

Gamification System

achievements

badges

user_badges

xp_history

streaks

leaderboards

Community System

posts

comments

likes

study_groups

group_members

AI System

chat_sessions

chat_messages

ai_recommendations

personalized_paths

Teacher System

teachers

classes

class_students

assignments

assignment_submissions

reports

Notification System

notifications

Field	Type
id	UUID
user_id	UUID
type	TEXT
title	TEXT
message	TEXT
read	BOOLEAN

Analytics Tables

events

page_views

user_sessions

simulation_usage

learning_analytics

Storage Architecture

avatars/

lesson-images/

lesson-videos/

3d-models/

simulation-assets/

documents/

certificates/

uploads/

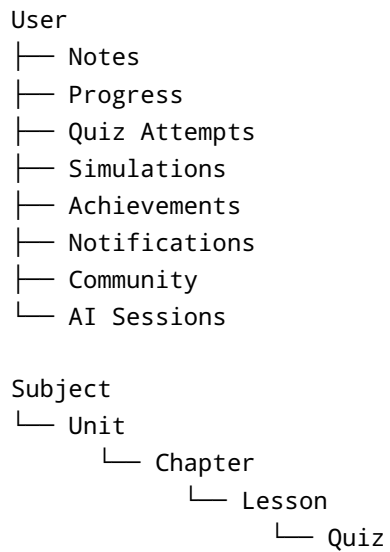
Row Level Security

Students: Only access their own data.

Teachers: Access assigned classes.

Admins: Full access.

Entity Relationship Overview



Future Scaling

Redis Cache

Search Engine

Event Queue

Microservices

Recommendation Engine

Data Warehouse

Machine Learning Pipeline

Final Goal

A database capable of supporting millions of learners and thousands of interactive simulations.